

Resolution Geometry

Stratified Specialization:

Continuation of the Observable Geometry across Walls of Distinguishability

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Abstract

Inside a stratum of constant observational rank, RG transitions are invertible and carry ordinary holonomy. At a wall where distinguishability is lost they are replaced by *specialization*, a non-invertible, information-collapsing passage to a lower stratum. This paper proves when each layer of the RG object specializes, and gives a counterexample for each hypothesis.

Under locally uniform convergence of the laws:

- the kernel groupoid is upper semicontinuous, $\limsup \mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$;
- a specialization *map* of quotients exists iff $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$, as for every garbling wall; otherwise specialization is a correspondence;
- specializations compose;
- regular symmetries descend exactly through the normalizer of $\mathcal{K}_0$, and the wall symmetry can be strictly larger;
- under C^1 convergence, tangent kernels are upper semicontinuous and rank is lower semicontinuous.

The distinguishability layer (Hellinger distance, Fisher metric, decision risk) specializes only under Hellinger continuity, that is, Le Cam convergence of the experiments. Weak convergence of the laws is not enough: a family converging weakly to a wall that identifies y keeps a constant Fisher metric and a minimal test error of 0.31 instead of 1/2. Locally uniform convergence cannot be certified by sampling and must be declared with a modulus. A rank wall is never certified from finite data. Specialization selects no latent connection.

A locked battery verifies every statement on computed families, with no hard-coded limits. It passes 11/11 on five seeds, and five code-level mutation canaries are caught.

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1 Walls and families

Two kinds of wall occur.

Protocol walls. A family of joint laws $F_\varepsilon : X \rightarrow Y$, $\varepsilon \in [0, \varepsilon_0]$, in which the protocol degenerates as $\varepsilon \rightarrow 0$ (a gain, a noise level, a bandwidth).

Latent walls. One law F whose differential rank changes across a closed subset of a regular stratum (L15 of the ladder [1]).

Write $\mathcal{K}_\varepsilon$ for the kernel groupoid of F_ε , Q_ε for its quotient and $V_\varepsilon = \ker DF_\varepsilon$. The law space Y carries the topology in which equality of laws is tested; statements about distinguishability carry the Hellinger metric H .

2 The exact layer

Theorem 2.1 (Kernel upper semicontinuity). *If $F_\varepsilon \rightarrow F_0$ locally uniformly and F_0 is continuous, then the Kuratowski upper limit of the kernel relations is contained in the wall kernel:*

$$\limsup_{\varepsilon \rightarrow 0} \mathcal{K}_\varepsilon \subseteq \mathcal{K}_0.$$

Proof. Let $(x_\varepsilon, x'_\varepsilon) \in \mathcal{K}_\varepsilon$ converge to (x, x') . Then $F_0(x) - F_0(x') = \lim[F_\varepsilon(x_\varepsilon) - F_\varepsilon(x'_\varepsilon)] + \lim[(F_0 - F_\varepsilon)(x_\varepsilon) - (F_0 - F_\varepsilon)(x'_\varepsilon)] + \lim[F_0(x_\varepsilon) - F_0(x)] - \lim[F_0(x'_\varepsilon) - F_0(x')] = 0$. The first bracket vanishes by membership, the second by local uniformity, the last two by continuity. $\square$

No-Go 2.2 (Pointwise convergence is not enough; SP02). Let $F_\varepsilon(x) = x + B_\varepsilon(x)$, with B_ε a bump of height 1 at $x = \varepsilon$ and width ε^2 . Then $F_\varepsilon \rightarrow \text{id}$ pointwise, but $(\varepsilon, 1 + \varepsilon) \in \mathcal{K}_\varepsilon$ converges to $(0, 1) \notin \mathcal{K}_0$ (separation 1.0). With bump height ε , which gives uniform convergence, the surviving pair converges into $\mathcal{K}_0$.

No-Go 2.3 (Uniformity is not certified by sampling; SP02). A sampling grid of spacing 10^{-3} reports sup-norm 0 for the bump family of No-Go 2.2. Locally uniform convergence is therefore a declared hypothesis H-unif, with a modulus such as a Lipschitz bound L , checked at every scale the data resolve.

- Without the declaration the certifier abstains.
- With $L = 10$ declared, it refuses the bump family, whose visible slope exceeds L .
- A certifier that trusts the grid sup-norm specializes the bump family (canary).

Proposition 2.4 (Specialization map; SP03). *1. A map $\sigma_\varepsilon : Q_\varepsilon \rightarrow Q_0$ with $\pi_0 = \sigma_\varepsilon \circ \pi_\varepsilon$ exists iff $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$; it is then unique and surjective.*

2. Every garbling wall $F_0 = M \circ F_\varepsilon$ has $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$ (L4); verified on 100/100 random merges.

3. Without inclusion, specialization is the correspondence $\limsup \mathcal{K}_\varepsilon$.

For the third case, $F_\varepsilon(x) = x^2 + \varepsilon x$ identifies $(x, -x - \varepsilon)$, which the wall $F_0(x) = x^2$ separates, so no map exists. The limit relation $\{(x, -x)\} \subseteq \mathcal{K}_0$ survives.

Proof. (1) is universal factorization (L2). (2) is L4. (3) is Theorem 2.1. $\square$

Proposition 2.5 (Functoriality and descent; SP04). 1. If $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_\delta \subseteq \mathcal{K}_0$, then $\sigma_{\delta 0} \circ \sigma_{\varepsilon \delta} = \sigma_{\varepsilon 0}$; this held in 100/100 computed chains.

2. A function on Q_ε descends along σ iff it is constant on $\mathcal{K}_0$ -classes.

A map that is not the factorization fails commutation (100/100 detected). A nearest-law matching used as the intermediate map is caught by the canary.

Proof. Both sides of (1) factor π_0 through π_ε , and the factorization is unique. (2) is universal factorization for $\mathcal{K}_0$. $\square$

3 The symmetry layer

Theorem 3.1 (Symmetry descent; SP05). An element $(\varphi, \psi) \in \text{Stab}(F_\varepsilon)$ induces a well-defined map of Q_0 iff $(\varphi \times \varphi)(\mathcal{K}_0) = \mathcal{K}_0$. The descended elements form a group whose image in $\text{Aut}(Q_0)$ can be a proper subgroup of the wall effective symmetry Γ_0 .

Proof. $[x]_0 \mapsto [\varphi x]_0$ is well defined iff φ maps $\mathcal{K}_0$ -classes to $\mathcal{K}_0$ -classes. Normalizers form a group. $\square$

The battery computes this by brute force over S_6 acting on six latent points and S_3 acting on outcomes. A regular symmetry that crosses wall classes fails to descend, so 1 of 2 regular symmetries descends. Meanwhile the wall group contains a cyclic $\mathbb{Z}_3$, so $|\Gamma_0| = 3$ against a descended image of order 1. Distinctions lost at the wall no longer break symmetry. Descending without the normalizer test produces an ill-defined action (canary).

4 The differential layer

Theorem 4.1 (Tangent upper semicontinuity; SP01). If $F_\varepsilon \rightarrow F_0$ in C^1 near x , then every Grassmann limit point of $V_\varepsilon(x_\varepsilon)$ with $x_\varepsilon \rightarrow x$ lies in $V_0(x)$, and $\text{rank } DF_0(x) \leq \liminf \text{rank } DF_\varepsilon(x_\varepsilon)$. The same holds within one law along $x_n \rightarrow x$ (latent walls).

Proof. Unit vectors $v_\varepsilon \in V_\varepsilon(x_\varepsilon)$ converging to v satisfy $DF_0(x)v = \lim DF_\varepsilon(x_\varepsilon)v_\varepsilon = 0$. Rank is lower semicontinuous because nonvanishing minors persist under small perturbations. $\square$

In 200 random families $A_\varepsilon = A_0 + \varepsilon B$, with rotating kernels of dimension $n - m$ converging into $\ker A_0$ of dimension $n - r$, and with nonlinear factor $\tanh$:

- the inclusion defect decays linearly (median order 1.000);
- the defect is $\leq 8.5 \times 10^{-6}$ at $\varepsilon = 10^{-6}$;
- rank is lower semicontinuous in every family.

Proposition 4.2 (Latent walls of a cusp; SP07). For $F(u, v) = (u, v^3 + uv)$ the computed strata are: two regular components; the fold $u = -3v^2$ with kernel transverse to it; and one cusp point, where the kernel becomes tangent to the fold. Along 300 random approaches to the fold, and along the fold towards the cusp, the kernel limits lie in the wall kernel (defect 0). For tame laws the stratification is finite [4, 7].

5 The distinguishability layer

Hypothesis 5.1 (H-hell). The family is locally uniformly Hellinger-continuous: $H(F_\varepsilon(x), F_0(x)) \rightarrow 0$ uniformly on compacta. For a dominated regular family with uniformly differentiable square-root densities this is the Le Cam convergence of the local experiments [5, 6].

Theorem 5.2 (Metric and decision specialization; SP06, SP08, SP11). *Under H-unif and H-hell:*

1. *Hellinger distances converge, $H(F_\varepsilon(x), F_\varepsilon(x')) \rightarrow H(F_0(x), F_0(x'))$; in particular pairs in $\mathcal{K}_0$ become indistinguishable;*
2. *natural Fisher modules converge, $g_\varepsilon \rightarrow g_0$, with $\text{rad } g_0 \supseteq \limsup V_\varepsilon$;*
3. *the Cramér–Rao bound diverges exactly on the gained kernel $V_0 \ominus \lim V_\varepsilon$;*
4. *Bayes risks of fixed decision problems converge to their wall values.*

Proof. (1) Triangle inequality for H . (2) Fisher information is the second-order Hellinger expansion, so uniform differentiability carries the limit. (3) Inversion of g_ε on a direction whose information tends to 0. (4) Risks are continuous in total variation, which is bounded by Hellinger distance. $\square$

No-Go 5.3 (Weak convergence does not carry distinguishability; SP06, SP11). Observe y as $\mathcal{N}(\varepsilon y, \varepsilon^2)$. As $\varepsilon \rightarrow 0$ the laws converge weakly to δ_0 , a wall that identifies all y (distance to the wall law $< 10^{-2}$). Yet:

- the Fisher information in y stays 1;
- $H(y = 0.3, y' = 1.1) = 0.277$ for every ε ;
- the minimal error for testing $y = 0$ against $y = 1$ stays 0.309, not the wall value $1/2$.

With $\mathcal{N}(\varepsilon y, 1)$ instead, Fisher $\rightarrow 0$, the Cramér–Rao bound grows as ε^{-2} (10^6 at $\varepsilon = 10^{-3}$), and the error tends to $1/2$. A gate based on weak convergence accepts the first family (canary).

In 200 linear-Gaussian families with fixed covariance:

- the Fisher metric converges with Lipschitz constant ≤ 28.6 in ε ;
- $\text{rad } g_0 = \ker A_0$ to defect 0;
- the Cramér–Rao bound on the gained kernel grows by a median factor 1.0×10^4 when ε shrinks by 100.

6 The typed stratified category

Definition 6.1 (Stratified quotient category). The objects are the quotients Q_α of the strata. The arrows are of two types:

- *regular* arrows: invertible transitions inside a stratum (the transition groupoid of the atlas [2]);
- *specialization* arrows: the maps σ of Proposition 2.4, or, without inclusion, the correspondences $\limsup \mathcal{K}$.

A path is admissible if every edge is admissible for its type.

Theorem 6.2 (Structure). *Under H-unif:*

1. *composition is associative, and specializations compose by Proposition 2.5;*

2. a specialization arrow is invertible iff $\mathcal{K}_\varepsilon = \mathcal{K}_0$;

3. holonomy is defined on the groupoid of each stratum. Two admissible paths into a wall are compared after specialization, $\sigma \circ P_1$ against $\sigma \circ P_2$. A difference that lies in the gained kernel is invisible, and any other difference is genuine stratified path dependence.

Proof. (1) Composition of maps and of closed relations. (2) σ is injective iff no new identifications occur. (3) Regular holonomy uses invertible arrows only. After specialization, endpoints are compared in Q_0 , where differences in $\mathcal{K}_0$ vanish by definition. $\square$

No-Go 6.3 (No connection from specialization; SP09). Horizontal lifts $(1, a)$ and $(1, b)$ of the same observable direction, differing by a vertical shear, give identical laws, identical Fisher modules (difference 0) and identical specializations, while the lifts differ by more than 3.5. Specialization selects no latent connection, so L58 is independent of it.

7 Statistical wall certificates

Proposition 7.1 (Rank certificates; SP10). *From finite data a rank wall is never certified.*

- Certificate *FULL* is issued if the 1% quantile of $\sigma_{\min}$ exceeds τ at sample sizes n and $4n$.
- Otherwise only an upper bound on $\sigma_{\min}$ is reported.

With $\tau = 0.005$:

<i>true $\sigma_{\min}$</i>	<i>certificate</i>	<i>bound</i>
0.08	<i>FULL</i>	—
0.004 (<i>true rank 2</i>)	<i>UPPER-BOUND</i>	≤ 0.0052
0 (<i>true wall</i>)	<i>UPPER-BOUND</i>	≤ 0.0015

Certifying a rank drop whenever the bound is below 2τ certifies the true rank-2 case as a wall (canary). This is the rank analogue of the zero-entropy lemma [3].

8 The stratified specialization theorem

Theorem 8.1 (Stratified specialization, locked form). *Let $F_\varepsilon \rightarrow F_0$ be a family of joint laws (or let $x_n \rightarrow x$ approach a latent wall of one law). Assume *H-unif*, a declared uniform convergence with a modulus. Then:*

1. exact: $\limsup \mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$. A specialization map exists iff $\mathcal{K}_\varepsilon \subseteq \mathcal{K}_0$, and is then unique, surjective and functorial. Exactly the $\mathcal{K}_0$ -basic invariants survive.
2. symmetry: regular symmetries descend iff they normalize $\mathcal{K}_0$; the wall group may be larger.
3. differential (with C^1 convergence): tangent kernels are upper semicontinuous and rank is lower semicontinuous; for tame laws the strata are finite.
4. distinguishability (with *H-hell*): Hellinger distances, natural Fisher modules and Bayes risks converge, and the Cramér–Rao bound diverges on the gained kernel.
5. global: the strata and specializations form the typed category of Theorem 6.2, with holonomy inside strata and post-specialization comparison across walls.

Each hypothesis is necessary:

- pointwise instead of uniform convergence breaks (1) (*No-Go 2.2*);

- inclusion is necessary for a map (Proposition 2.4);
- the normalizer condition is necessary for (2) (Theorem 3.1);
- Hellinger rather than weak convergence is necessary for (4) (No-Go 5.3).

Not implied: an isomorphism of quotients, preservation of dimension, survival of a nondegenerate metric, or a canonical latent connection (No-Go 6.3).

9 Ladder insertion

Rung	Statement	Status	Ref.
L16.1	Kernel upper semicontinuity under H-unif; pointwise is not enough; uniformity is declared	forced (no-go)	Thm. 2.1
L16.2	Specialization map iff kernel inclusion; else correspondence; functorial	forced	Prop. 2.4
L16.3	Symmetry descent through the normalizer of $\mathcal{K}_0$; wall group may grow	forced	Thm. 3.1
L16.4	Tangent upper semicontinuity, rank lower semicontinuity; finite tame strata	forced / imported	Thm. 4.1
L16.5	Distinguishability specializes iff Hellinger (Le Cam) convergence	hypothesis H-hell	Thm. 5.2
L16.6	Typed stratified category; holonomy within strata; post-specialization comparison	forced	Thm. 6.2
L16.7	Rank walls never certified from finite data	forced (statistical)	Prop. 7.1
L16.8	Specialization selects no connection (L58 independent)	no-go	No-Go 6.3

A Reproducibility

`rg_stratified_specialization_attack.py` passes 11/11 on seeds 20261001, 1, 42, 777 and 2026. `--canary` mutates code paths and requires the targeted check to fail:

- trust the grid sup-norm (SP02);
- use nearest-law matching as the intermediate map (SP04);
- descend without the normalizer test (SP05);
- use a weak-convergence metric gate (SP06);
- certify a rank drop from a small bound (SP10).

All five are caught. Corrections before registration:

1. SP01 first used unscaled base points, whose tanh saturation created numerical kernel directions.
2. SP05 first used a regular symmetry that preserved the wall partition, so it could not reject anything.
3. SP02 first certified uniformity from a grid sup-norm, which is blind to narrow bumps; it was recast as No-Go 2.3.

B Disposition of the earlier battery SC01–SC12

Replacing the inclusion test of the earlier battery by the constant 0 still passed 12/12. Its kernel limits and walls were hard-coded (SC01, SC02). SC04 compared $2x + 1$ with itself. Its canary (SC12) toggled a boolean in a dictionary and never touched the other checks.

- Its structural claims are retained here, now computed and locked: SC01/SC02 $\rightarrow$ SP01; SC04 $\rightarrow$ SP04; SC05 $\rightarrow$ SP04; SC06 $\rightarrow$ SP05; SC07 $\rightarrow$ SP06/SP08; SC08 $\rightarrow$ SP09; SC09/SC10 $\rightarrow$ Theorem 6.2; SC11 $\rightarrow$ SP10.
- The metric gate is corrected. Divergence of an inverse metric (SC07) is the Cramér–Rao signature of a correct specialization, not a failure. The real failure is weak convergence without Hellinger convergence.

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